

Part A — Conceptual Understanding (Theory)

Q1. What are Supervised Learning Algorithms?

Supervised learning algorithms learn from labeled training data — where both input features and the correct output are provided. The model learns the mapping from inputs to outputs and uses it to make predictions on new, unseen data. Examples: Linear Regression, Decision Tree, KNN.

Q2. Difference between Regression and Classification Algorithms

Regression predicts a continuous numeric value (e.g., house price, salary).

Classification predicts a discrete category or label (e.g., spam/not spam, disease/no disease).

Example: Predicting house price → Regression. Predicting if a house will sell → Classification.

Q3. Explain Simple Linear Regression

Simple Linear Regression models the relationship between one independent variable (X)

and one dependent variable (Y) using a straight line: $Y = mX + c$

where m is the slope (how much Y changes per unit of X) and c is the intercept (value of Y when $X = 0$).

Q4. Assumptions of Linear Regression

1. Linearity — Relationship between X and Y is linear.

2. Independence — Observations are independent of each other.
3. Homoscedasticity — Residuals have constant variance across all values of X.
4. Normality of Residual — Residuals (errors) are normally distributed.
5. No Multicollinearity — Independent variables are not highly correlated with each other.

Q5. Bias–Variance Trade-Off

Bias is the error from overly simple models that miss patterns (underfitting).

Variance is the error from overly complex models that learn noise (overfitting).

The trade-off means reducing bias often increases variance and vice versa.

The goal is to find a model that balances both — low bias and low variance — for best generalization.

Q6. Overfitting and Underfitting

Underfitting occurs when a model is too simple to capture the data pattern —

it performs poorly on both training and test data. Example: fitting a straight line to curved data.

Overfitting occurs when a model is too complex and memorizes training data including noise —

it performs well on training data but poorly on test data. Example: a degree-30 polynomial on simple data.